

Ann Biju Mariyam

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Results-driven Data Analyst with hands-on experience in data analysis, statistical analysis, data visualization, and dashboard development across retail, e-commerce, and public health domains. Proficient in SQL (JOINS, aggregations, window functions), Python (Pandas, NumPy), Tableau, and Power BI to generate business insights that support data-driven decision-making and stakeholder reporting. Skilled in data cleaning, data validation, data pipeline development, ETL pipelines, time series forecasting, and KPI development. Implemented and evaluated machine learning models using Scikit-learn and TensorFlow with a focus on real-world deployment impact.

PROFESSIONAL EXPERIENCE

Data Analytics Intern, ROGERSOFT TECHNOLOGIES, India

Sep 2024 – Dec 2024

Project: Rogersoft Technologies' business and product teams relied on accurate, analysis-ready datasets to track product performance, customer behavior, and operational metrics across multiple internal systems. Analytics and business stakeholders relied on consistent reporting and self-service dashboards to monitor KPIs, identify trends, and make data-driven decisions without the overhead of manual reporting.

Objective: Support analytics and reporting workflows by developing SQL-based data solutions, improving data quality and validation processes, and building interactive dashboards that enabled reliable, self-service insights for cross-functional business teams.

- Queried and developed advanced SQL solutions (multi-table JOINS, GROUP BY, conditional filters) to analyze product and customer datasets, translating raw data into performance metrics for data-driven stakeholder reporting and reduced manual analysis overhead.
- Streamlined data cleaning, validation, and quality checks on 10,000+ records using Python (Pandas) and Excel, producing analysis-ready datasets that reduced reporting inconsistencies and improved pipeline efficiency; designed interactive Power BI dashboards with KPI cards, trend lines, and bar charts, enabling teams to monitor cross-functional operational metrics and act on findings independently.

PROJECTS

Sales Revenue Analytics [Github](#)

Analyzed retail sales transaction data and built an end-to-end analytics workflow using Python, SQL, and BI tools to diagnose profitability gaps and deliver actionable revenue insights for data-driven decision-making.

- Analyzed 9,994 transactions (~\$2.3M revenue) using **Python** (Pandas, Matplotlib, Seaborn) and **SQL** to diagnose a **profitability gap**, pinpointing Technology as the top profit category (\$145K), Tables as the largest loss driver (-\$17.7K), and discounts above 30% as a margin risk across product lines.
- Delivered interactive **Tableau** and **Power BI dashboards** with **KPI** tracking and monthly revenue trends, plus a **stakeholder-ready** report documenting findings and recommended optimizations, enabling leadership to act on data-driven insights.

E-commerce Customer Analytics & ETL [Github](#)

Built an end-to-end ETL pipeline and analytics workflow on the Olist e-commerce dataset using Python and SQL to unify multi-table relational data and deliver customer experience, logistics, and retention insights for operational decision-making.

- Developed an **ETL pipeline** using **Python (Pandas, NumPy)** and **SQL** integrating 6 relational tables (~98,666 orders, ~\$15.84M revenue) into a unified reporting layer, enabling consistent analysis across logistics, retention, and customer experience functions.
- Surfaced that longer **delivery times** correlate with lower review scores and that **freight costs** are disproportionately high in key states, delivering **Power BI dashboards** and a prioritized operational improvement roadmap backed by quantitative statistical analysis and **KPI** benchmarking.

COVID-19 Regional Forecasting - Italy (MS Capstone) [Github](#)

Built an end-to-end time series forecasting pipeline on Lombardia regional clinical epidemiological data using Python, evaluating and comparing statistical and machine learning models to support public health resource planning and regional health authority decision-making.

- Developed and benchmarked **Naive Baseline, SARIMA, and Random Forest forecasting models** using **Python (Pandas, Scikit-learn)**, with SARIMA achieving the lowest forecasting error (**MAE: 180** vs. **Naive MAE: 206**); documented business and functional findings in stakeholder-ready reports.
- Identified and corrected a critical model evaluation flaw through rigorous statistical validation and data integrity checks, ensuring analytical outputs met research-grade accuracy standards for downstream reporting and compliance.

SKILLS & CERTIFICATIONS

Skills:

Programming Languages: Python (NumPy, Pandas, Matplotlib, Scikit-learn), SQL, R Programming
Analytics: Exploratory Data Analysis (EDA), Data Summarization, Statistical Analysis, Statistical Modelling, Hypothesis Testing, Trend Analysis, Data Cleaning, Data Validation, Data Quality, ETL Pipelines, Time Series Forecasting, KPI Analysis, Stakeholder Reporting
Databases : MySQL, Microsoft SQL Server, Relational Database Design
Visualization & Reporting: Power BI (Power Query, DAX, Dashboard, KPI Cards), Tableau, Excel(Pivot Tables, XLOOKUP, Advanced Analysis), Looker
Tools: GitHub, Jupyter Notebook, Google Colab
Machine Learning & Analytics: Supervised ML, Unsupervised ML

Certifications:

Microsoft Power BI Data Analyst Associate (PL-300), Google Cloud Data Analytics.

EDUCATION

UNIVERSITY OF ILLINOIS SPRINGFIELD <i>MS in Data Analytics</i> (GPA: 3.9/4)	Springfield, IL, USA May 2026
FEDERAL INSTITUTE OF SCIENCE AND TECHNOLOGY <i>BTECH in Computer Science and Engineering</i>	Kerala, India June 2024